

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) is now a central framework for strengthening the reasoning capabilities of large language models. Yet widely used methods such as GRPO leave substantial learning signal unused because zero gradients arise from fixed token-level clipping bounds and from standardizing identical rewards within the current step. These mechanisms produce ineffective updates and underutilize generated responses precisely when RLVR needs efficient exploration. We present **Dynamic Clipping Policy Optimization (DCPO)**, a policy-optimization method that directly resolves these two bottlenecks: dynamic clipping adapts the probability-ratio bounds to token-specific prior probabilities to expand token-level exploration where it is most needed, while smooth advantage standardization accumulates reward statistics across training steps to recover response-level learning signal from generated data. Across four benchmarks and four model backbones, DCPO delivers state-of-the-art performance. On Qwen2.5-Math-7B, DCPO reaches 46.7 Avg@1 under greedy decoding and 38.8 Avg@32 under 32-time sampling on AIME24, outperforming DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9). On Qwen2.5-14B, DCPO achieves 23.3/19.0 on AIME25, exceeding GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). DCPO also improves the nonzero advantage ratio by an average of 28% over GRPO across four models, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude compared with GRPO and DAPO while achieving stronger accuracy. These results establish DCPO as a more data-efficient and exploration-preserving RLVR optimizer for large language models.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a primary driver of recent progress (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a particularly effective route for improving these capabilities because it optimizes LLMs with rule-based outcome rewards. This verifiable supervision directly targets domains that demand reflective decision-making, including mathematics and coding (DeepSeek-AI et al., 2024).

The GRPO family of algorithms (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) provides a strong foundation for RLVR, and its effectiveness has been validated across multiple studies. However, this foundation also exposes a clear optimization bottleneck. Recent work, including DAPO (Yu et al., 2025), shows that GRPO can suffer from entropy collapse, with advantage values decaying to zero in later training stages. In addition, fixed and inappropriate weight clipping discards learning signals from rarely sampled tokens, unnecessarily narrowing the model’s exploration space.

DAPO addresses part of this problem through Clip-Higher, which relaxes the upper clipping bound to mitigate entropy collapse, and Dynamic Sampling, which filters uniformly rewarded samples so each batch contains valid gradients. These design choices improve over vanilla GRPO, but they do not remove two core inefficiencies. First, **Response-Level Inefficiency**: dynamic sampling trades gradient validity for sampling efficiency and therefore slows convergence. Second, **token clipping deficit**: our analysis shows that the appropriate clipping bound is token-dependent, so any single uniform upper bound remains structurally suboptimal. Methods that avoid directly discarding zero-

gradient responses still inherit these limitations unless they address both response-level utilization and token-level clipping.

We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a direct solution to these optimization failures. First, DCPO uses a **smoothing technique that standardizes advantages across cumulative steps**, preventing responses with identical rewards in the current step from being reduced to zero-advantage updates. By aggregating the reward distribution of cumulatively generated responses with the current distribution, DCPO preserves usable response-level exploration signal (see Section 2). Second, DCPO replaces fixed clipping with a **dynamic clipping method that adaptively adjusts the clipping bounds based on the old token-specific probabilities**. Rather than selecting another pair of fixed upper and lower thresholds, DCPO grants wider bounds to tokens with lower prior probabilities, thereby creating principled space for token-level exploration (see Section 2 and Appendix A.6). Finally, DCPO computes the mean loss only across tokens within each individual response instead of averaging over the full batch, preserving the relative advantage structure among responses to the same prompt and avoiding the dilution introduced by batch-level averaging.

We evaluate this design on a merged training set combining Math-DAPO-17k (Yu et al., 2025) with levels 3-5 of the MATH dataset (Liu et al., 2025), and we test on four standard mathematical reasoning benchmarks. Experiments across four models against GRPO, DAPO, and, where available, GSPO demonstrate that DCPO consistently delivers the strongest performance. The gains are especially pronounced on the challenging AIME benchmarks: on AIME24-Avg@32 with 32-time sampling, DCPO-7B scores 38.8 versus 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO), while on AIME25-Avg@32, DCPO-14B reaches 19.0 versus 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). Beyond accuracy, DCPO raises the average nonzero advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude compared with the baselines.

Appendix A.2 provides the preliminary loss and advantage calculations for prior algorithms, including GRPO, DAPO, and GSPO.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Probability-ratio clipping is a core stabilization mechanism in reinforcement learning because it prevents policy optimization from making overly large parameter shifts. Its role becomes decisive in off-policy importance sampling. Reweighting by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$ yields an unbiased estimate of $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$, but it can also sharply increase variance, as formalized in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

Prior work, including Schulman et al. (2017), addresses this variance by constraining the probability ratio with a fixed clipping hyperparameter ϵ , as shown in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping bound is too rigid for RLVR because it constrains exploration most severely where the old policy assigns low probability mass. In these regions, the allowable decrease is inherently narrow, so meaningful policy updates can be suppressed. Asymmetric clipping in methods such as DAPO relaxes this restriction but still applies a global bound that cannot reflect token-level probability differences. GSPO further exposes the cost of coarse clipping: when the sequence-level importance ratio exceeds the asymmetric bounds, it discards entire responses, wasting informative tokens while retaining high-variance tokens that can destabilize training.

A stronger variance–bias tradeoff is obtained by imposing probability-dependent constraints rather than fixed global ones, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3), and enforcing non-negativity of the square-root term by taking its maximum with zero, gives a closed-form solution for $r(x)$. The resulting **dynamic-**

adaptive clipping bounds depend explicitly on $q(x)$ and on the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

Dynamic-adaptive clipping therefore expands policy exploration exactly in low-probability regions where fixed bounds are most restrictive. Following the motivation of dual clipping (Ye et al., 2020), we cap the maximum bounds for both positive and negative advantages at 10 to prevent excessively wide bounds from causing overtraining. Appendix A.5 and Appendix A.6 provide the full derivation and analysis from the importance-sampling perspective.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

Prior methods such as GRPO, DAPO, and GSPO compute the advantage $\hat{A}_{new,j}^i$ of the j -th response by standardizing only the rewards $R_{j=1,\dots,G}^i$ for the same prompt at step i , as shown by Equation (11) in Appendix A.3. This step-local standardization creates two concrete failure modes. First, when sampling randomness makes all rewards identical in a step, every advantage becomes zero, and the prompt contributes no parameter update even if the generated reasoning trajectories contain useful differences. Second, high-entropy sampling can produce skewed label counts, causing standardized advantages to fluctuate sharply across steps and even reverse sign, which destabilizes training.

Recent evidence (Yue et al., 2025) further indicates that RLVR primarily refines a policy by adjusting the likelihood of correct samples rather than replacing the underlying distribution. This view makes the reward distribution for responses to the same prompt a natural training-wide object, motivating *cumulative* standardization as defined in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics aggregate all responses generated so far for the same prompt.

To control fluctuations from both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), we introduce two smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$. These functions form weighted averages of the two standardization schemes, with weights that change over step i as shown in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

$\hat{S}A_{new,j}^i$ increasingly emphasizes the current distribution as training proceeds, enabling the optimizer to adapt to the latest policy outputs. Conversely, $\hat{S}A_{total,j}^i$ increasingly emphasizes the cumulative distribution, treating the current-step deviation as a fine adjustment within the broader reward history. DCPO then defines the final advantage $\hat{A}_j^i$ as the smoothed advantage with the smaller absolute value, which directly limits the instability caused by either cumulative or current-step fluctuations, as specified in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

Once a prompt enters optimization, responses generated in subsequent steps continue to contribute to model updates even when the current step would otherwise assign zero advantage. Thus, when identical rewards occur within a step, these responses still update the model through the advantage $\frac{1}{i} \hat{A}_{total,j}^i$, preserving useful learning signal and improving data efficiency.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)** (the loss Equation (13) used in DAPO) balances token contributions, but it also distorts the relative importance of responses with different lengths. Consider two responses to the same prompt: response A has advantage 1 and length 500, while response B has advantage 0.5 and length 1500. Under TLM, A receives weight $\frac{500}{500+1500} = 0.25$, and B receives weight $\frac{1500}{500+1500} = 0.75$. The total loss contribution is therefore $1 \times 0.25 = 0.25$ for A and $0.5 \times 0.75 = 0.375$ for B , so the lower-advantage response exerts the larger influence.

The **Sequence-Level Mean loss (SLM)** (the loss Equation (12) used in GRPO) introduces the opposite problem: averaging across the batch weakens the advantage magnitude, effectively reducing it by G after advantage calculation. With large batches, this averaging can damage the standardized relative structure among responses. DCPO treats standardized advantages as already encoding the response-level relationships, and therefore averages only across tokens within the same response using $|o_i|$. This preserves the original relative ordering while distributing each response’s importance equally across its own tokens. We call this formulation **Only Token Mean loss (OTM)**, defined in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, DCPO does not constrain this loss with KL divergence. The lower and upper clipping bounds are the dynamic-adaptive bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets We evaluate DCPO on four representative model backbones: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). Training uses a merged set of approximately 25k mathematical problems, combining the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). We process every problem with the Qwen-Math template (see Section A.8) for both training and evaluation, ensuring a consistent input format across all comparisons.

Evaluation We evaluate on four established mathematical reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, its larger size (500 problems) makes Avg@1 the primary reported metric. For the smaller and more sampling-sensitive **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) datasets, we report both Avg@1 and Avg@32 to measure deterministic accuracy and distributional robustness.

- **Avg@1:** This metric reports standard greedy-decoding accuracy and captures the model’s strongest single prediction.
- **Avg@32:** This metric averages the accuracy of 32 sampled responses per problem with temperature 1.0 and top_p 1.0, directly measuring the robustness and stability of the trained policy distribution.

Baseline Settings We compare against GRPO and DAPO across all model settings, and we include GSPO as an additional baseline on Qwen2.5-Math-7B and Qwen2.5-14B. Appendix A.9 specifies the remaining experimental settings, including clipping thresholds ϵ_* , codebase details, and common training hyperparameters.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 compares GRPO, DAPO, GSPO, and DCPO on MATH500, AMC23, AIME24, and AIME25. Reporting both greedy decoding (Avg@1) and sampling-based decoding (Avg@32) makes the comparison sensitive to both peak prediction accuracy and policy-distribution quality. Across

Table 1: Benchmark performance of DCPO and baselines. Avg@1 reports greedy decoding, Avg@32 reports 32-sample accuracy, boldface marks the best result, and Δ quantifies DCPO’s gain over each baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

model scales, DCPO is consistently the strongest or a highly competitive optimizer, establishing its effectiveness as a general-purpose RLVR policy-optimization method for LLMs.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO achieves the highest average scores (47.2/32.8), improving over GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), clearly exceeding GRPO (+1.3/+1.7) and remaining competitive with DAPO, with a +1.0 Avg@1 gain and only a -0.4 Avg@32 gap. On Qwen2.5-Math-7B, DCPO attains (57.1/45.2), outperforming GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On Qwen2.5-14B, the largest tested model, DCPO achieves (53.2/39.0), substantially improving over GRPO (+6.5/+7.7) and exceeding DAPO by (+1.3/+0.1) and GSPO by (+4.2/+5.5). These results show that DCPO’s advantages persist across model capacities and under both greedy and stochastic decoding.

Robustness on AIME under Sampling (Avg@32) The most decisive gains appear on AIME24 and AIME25, high-difficulty competition benchmarks that stress-test mathematical reasoning. Under Avg@32, which evaluates correct generation under high-entropy sampling, DCPO is markedly more robust at the 7B and 14B scales. On AIME24-Avg@32, DCPO-7B achieves 38.8, outperforming GRPO (32.1), DAPO (31.6), and GSPO (34.9). On AIME25-Avg@32, DCPO-14B reaches 19.0, exceeding GRPO (10.5), DAPO (15.3), and GSPO (9.9). These results indicate that DCPO improves

not only average performance but also the diversity and reliability of correct sampled generations, a crucial property when inference relies on multiple plausible outputs.

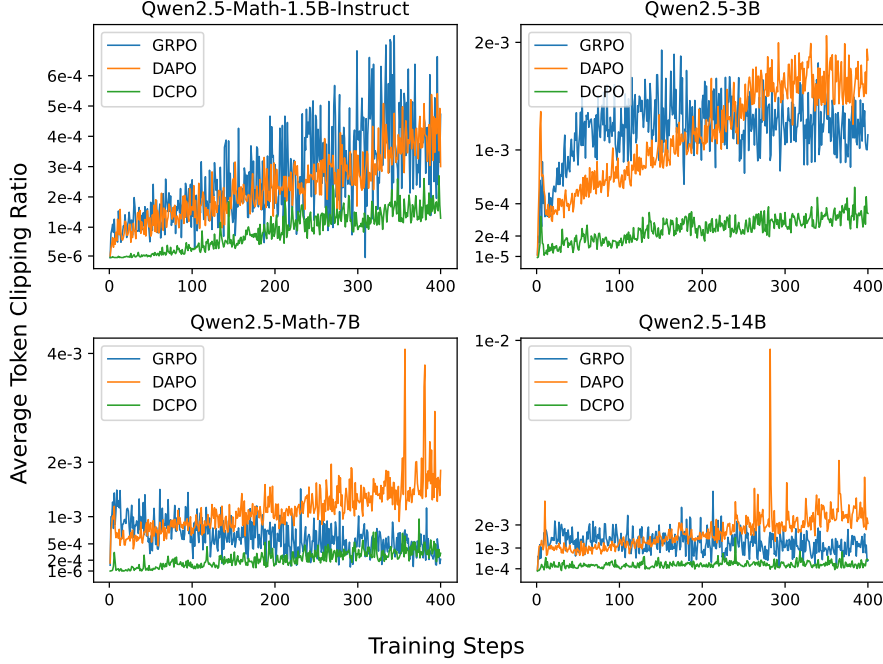


Figure 1: Token clipping ratio (TCR) across models and methods, showing DCPO’s substantially lower clipping burden.

5.2 THE TOKEN CLIPPING RATIO (TCR)

We define the **Token Clipping Ratio** (TCR) as the proportion of tokens excluded from policy updates during back-propagation because of clipping. Figure 1 shows that TCR sharply distinguishes DCPO from existing reinforcement learning methods.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N denotes the number of microbatches per training step.

GRPO shows model-size-dependent TCR dynamics: for Qwen2.5-Math-1.5B and Qwen2.5-3B, TCR rises with training, whereas for Qwen2.5-Math-7B and Qwen2.5-14B it gradually declines. DAPO is less favorable, showing a consistent upward TCR across all models; as training proceeds, more tokens are clipped and more updates are based on partial or truncated responses. Both GRPO and DAPO also exhibit large fluctuations and abnormal spikes, which can inject instability and contribute to entropy collapse. GSPO incurs an even heavier clipping burden: its average TCR exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B, substantially above token-level clipping methods and therefore discarding more informative tokens.

DCPO instead maintains a stable and efficient clipping profile. Across model sizes and training stages, DCPO’s TCR remains nearly constant and an order of magnitude lower than GRPO and DAPO. This lower, steadier TCR means DCPO discards fewer tokens, allows more complete response information to shape policy updates, and preserves exploration capacity for rare tokens while still controlling outlier variation.

Qwen2.5-Math-7B illustrates why adaptive clipping matters. The token distribution rapidly concentrates on high-confidence tokens: after roughly 60 steps, about 95% of generated tokens satisfy $q(x) > 0.9$; after roughly 100 steps, this fraction reaches about 97% and continues to rise. As shown

in Figure 4d, once the old probability satisfies $(q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83)$, the new probability $p(x)$ participates in model updates within $[\frac{1}{1+\epsilon}, 1]$ without being discarded by any clipping method. Thus, most tokens occupy a region that is never clipped, explaining the low overall *TCR* in practice. The later-stage *TCR* growth in Figure 1 therefore points to the important failure case: low-probability tokens are increasingly discarded, limiting rare-token exploration.

Previous work (Wang et al., 2025) identifies **high-entropy tokens**—the low-probability tokens in the policy distribution—as primary drivers of emergent reasoning capability. Because lower probability corresponds to higher entropy, rare tokens carry especially informative exploration signal. Granting these tokens a broader admissible update range therefore enlarges the model’s effective exploration space and supports more diverse reasoning behavior.

Together, these empirical and theoretical observations motivate **dynamic-adaptive clipping**: DCPO relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the safe interval for high-confidence tokens. By expanding the update interval specifically for the high-entropy tail, DCPO restores learning from rare informative tokens and improves reasoning performance.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a generated response contributes to policy optimization only when it produces a nonzero gradient. We quantify this data-efficiency bottleneck with the **Response Utilization Ratio** (*RUR*), the percentage of generated responses with nonzero advantage that participate in policy updates.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 summarizes the overall average *RUR* for responses generated during GRPO training is low, with an overall mean of only 44.6% across four different models. This indicates that more than half of the generated responses are discarded, highlighting the significant inefficiency in response utilization during the exploration of the diversity. In contrast, our proposed DCPO method achieves an average *RUR* of approximately 70% after the first epoch. This represents a substantial improvement, with an absolute increase of 28% and a relative improvement of 64% over the GRPO’s.

Table 2: Average response utilization ratio (*RUR*) over 400 training steps.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 further shows that GRPO’s *RUR* deteriorates over time across model sizes and model types. It falls from above 90% to below 50%, and on Qwen2.5-Math-7B it drops to roughly 30% after about 200 training steps. This pattern confirms that current-step advantage standardization creates a growing zero-gradient fraction and makes late-stage exploration inefficient. DCPO, by contrast, keeps *RUR* high through smooth advantage standardization: after the first epoch, *RUR* stabilizes around 70% and continues to improve with additional training.

The early *RUR* exceeds 90% for a specific and limited reason, while the later value stabilizes near 70% rather than approaching 100%. This behavior clarifies why SAS improves utilization without forcing every response into the update.

First, the initial *RUR* above 90% largely reflects early model outputs that fail to follow the required format, receive negative rewards, and inflate utilization. Once models quickly learn the fixed answer pattern, training must focus on improving reasoning rather than format compliance. Second, two factors explain why some zero-advantage responses remain and why *RUR* stabilizes near 70%: (i) many math problems are already consistently solved correctly, so excluding them from updates is appropriate because the relevant capability no longer requires refinement; (ii) other instances are consistently erroneous, either because the problems are too difficult for the current model even with a correct ground-truth label, or because the DAPO-Math-17K dataset contains items whose labels were converted into mislabeled integers, creating potentially “incorrect” labels. In these cases, discarding zero-advantage responses acts as useful filtering and reduces the risk of updating on noisy or mislabeled data.

DAPO discards generated responses with identical labels, so direct *RUR* comparison with GRPO and DCPO is not well defined. We therefore evaluate DAPO’s efficiency by measuring how many responses and how much training time are required to complete the same number of update steps. To offset the low sampling efficiency caused by GRPO’s low *RUR*, DAPO uses a generation batch size 1–4 times the training batch size and adjusts it dynamically. Across four models, DAPO requires 3 to 5 times more generated responses than DCPO for the same number of parameter updates, at least doubling GPU hours without producing better performance. Its utilization also worsens with model size: on Qwen2.5-14B, the proportion of responses used by DAPO falls below 30% after about 300 steps. These results show that dynamic sampling sacrifices substantial data utilization and leaves exploration potential unused.

Overall, DCPO advances beyond GRPO and DAPO by using substantially more generated responses, improving training efficiency, preserving exploration space for diverse reasoning trajectories, and translating these efficiency gains into stronger benchmark performance.

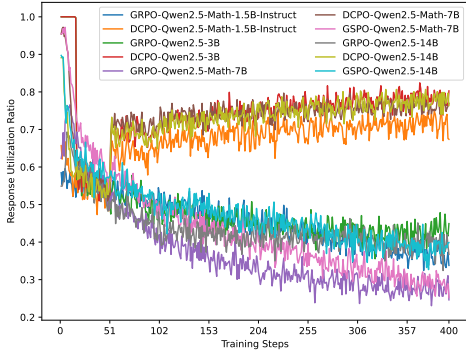


Figure 2: Response utilization ratio (*RUR*) during training.

5.4 ABLATION STUDIES

We isolate the contribution of each DCPO component on Qwen2.5-Math-7B using **Avg@32**, which directly reflects robustness and stability of the learned policy distribution. For a controlled comparison, each experiment modifies one component of the GRPO baseline while keeping all other settings fixed and removing the KL divergence term to match DAPO, GSPO, and full DCPO. Figure 3 reports the learning curves from 20 to 400 steps.

The component curves show sustained improvement in inference ability. Each individual modification surpasses GRPO and reaches or exceeds the performance of DAPO and GSPO, while the full combination delivers the largest gain. Baselines plateau in later training, whereas DCPO continues to improve both at the component level and as a complete method. The ablation therefore confirms that OTM, SAS, and DAC each provide a distinct optimization benefit and that their integration yields cumulative gains in stability and performance.

(i) **Only-Token-Mean loss (OTM)**: Aggregating loss within each response rather than across the whole batch already improves GRPO, showing that preserving response-level advantage structure is valuable. (ii) **Smoothed Advantage Standardization (SAS)**: Replacing stepwise advantage with cumulative smoothed advantage substantially improves GRPO and approaches DAPO, demonstrating that cross-step reward statistics stabilize optimization and improve final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: Replacing GRPO’s fixed clipping boundary with probability-adaptive clipping produces a major improvement; DAC alone surpasses GRPO, DAPO, and GSPO and is more stable than OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: Combining the three mechanisms achieves the strongest and most stable results, decisively outperforming GRPO, DAPO, and GSPO. The synergy among OTM, SAS, and DAC shows that each design choice targets a different source of inefficiency in RLVR training.

In sum, the ablation study verifies that every DCPO component contributes positive value and that their combination produces substantial cumulative gains. These results support DCPO’s central claim: data utilization, adaptive token exploration, and response-level loss aggregation must be addressed together for stable and efficient RLVR optimization.

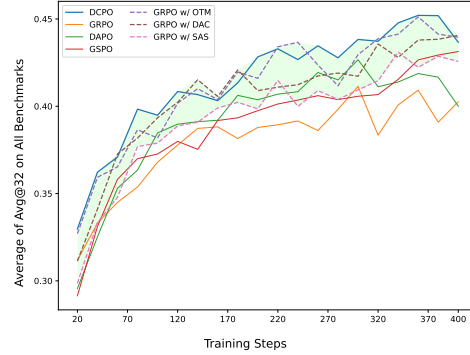


Figure 3: Component ablation on Qwen2.5-Math-7B using average Avg@32.

6 CONCLUSION

This work directly addresses two central limitations of existing RLVR policy optimizers: restricted token-level exploration caused by fixed clipping in low-probability regions, and low sample efficiency caused by current-step reward standardization. We introduced **Dynamic Clipping Policy Optimization (DCPO)**, a reinforcement learning pipeline that strengthens LLM reasoning through two coordinated mechanisms: (i) **dynamically adaptive token-level clipping**, which sets bounds from old policy probabilities and enables more effective exploration of rare low-probability tokens, and (ii) **Cumulative-Smooth Advantage Standardization**, which aggregates reward distributions across optimization steps and reduces reward randomness from high-entropy sampling. Experiments on four mathematical reasoning benchmarks and four models demonstrate that DCPO consistently improves performance and efficiency. DCPO-7B, for example, reaches 38.8 Avg@32 on AIME24, outperforming GRPO (32.1) and DAPO (31.6). DCPO also increases response utilization by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude. These findings establish DCPO as a more efficient and exploration-preserving RLVR optimizer. Future work will extend DCPO to additional domains, including code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

Existing methods, including GRPO and DAPO, compute the token advantage $\hat{A}_{j,t}^i$ for token t in response j by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the G responses generated at step i , as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When all responses to the same prompt receive identical rewards, this formulation assigns zero advantage and wastes otherwise generated responses because they no longer contribute to model updates.

A.3.1 GRPO

Shao et al. (2024) introduced **Group Relative Policy Optimization (GRPO)**, which samples G responses per query, assigns rewards R with a rule-based reward function, and estimates token-level advantages through Equation (11). The resulting parameter update uses Equation (12), the **Sequence Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages within each response and then across all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty limits deviation from a reference policy, such as a pre-trained model, while the fixed clipping threshold ε restricts large token-ratio changes. This standard averaging strategy stabilizes training but also dilutes per-response importance.

A.3.2 DAPO

Yu et al. (2025) proposed **Dynamic sAmpling Policy Optimization (DAPO)** to address GRPO’s tendency to let longer sequences dominate gradient updates. DAPO normalizes token counts across the full batch, yielding the **Token Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. This rule discards prompts whose responses all share the same reward and regenerates responses to maintain batch size, which improves gradient validity but creates severe data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) proposed **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping can introduce high-variance noise into gradient estimates. Because this noise can accumulate over long sequences and be amplified by clipping, GSPO replaces token-wise clipping with sequence-wise clipping to obtain a more stable training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp\left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log\left(\frac{\pi_\theta(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)}\right)\right)$$

GSPO nevertheless discards more than 10% of responses with nonzero advantages, which produces substantial data waste and reduces training efficiency. Conversely, GRPO does not remove many high-entropy token-level responses, leaving a source of instability that DCPO targets more directly.

A.4 REWARD CALCULATION

We use a rule-based reward model to provide supervision that is both verifiable and interpretable. As defined in Equation (15), each generated response is scored by answer correctness and format compliance: +1 for correct format and correct answer, 0 for correct format but incorrect answer, and -1 for incorrect format.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response group for a given problem at training step i , and the reward depends only on response format and correctness.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Probability-ratio clipping is essential for stable reinforcement learning because it prevents large, destabilizing policy updates. The need is especially clear under importance sampling, where samples from an old policy are reused to estimate expectations under a new policy.

Importance sampling rewrites the expected value of $f(x)$ under the new probability $p(x)$ as an expectation under the old probability $q(x)$, using the probability ratio as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although the estimator is unbiased, it can substantially inflate variance, the central challenge addressed by clipping. Equation (17) gives the true variance of $f(x)$ under the new policy, while Equation (18) gives the variance of the importance-sampled estimator under the old policy:

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The gap between the importance-sampled variance and the true variance makes the inflation explicit, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To control this variance, PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025) commonly impose a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, necessarily limits exploration in regions where the old policy assigns low probability mass. In these regions, the absolute update range is severely restricted, so meaningful updates to rare but informative tokens can be suppressed. DAPO’s asymmetric bounds relax the constraint, but they remain non-adaptive and therefore retain the core limitation.

Recent studies such as Yue et al. (2025) further suggest that the reasoning paths used by RLVR models are largely already present within the base model’s sampling distribution. The small fraction of clipped parameters during off-policy batch updates also indicates limited parameter shift. This stability motivates our probability-aware alternative: instead of the fixed clipping rule in Equation (20), DCPO constrains $r(x)$ through the dynamic-adaptive restriction in Equation (21).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the restriction yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For the clipping ratio to remain mathematically valid, the derivation must respect the basic constraints of probability distributions and probability ratios. As shown in Equation (23), $p(x)$ and $q(x)$ are bounded between 0 and 1, while $r(x)$ and the clipping hyperparameter ϵ are non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

These constraints lead to the necessary bound conditions in Equation (24), which make the dynamic clipping rule both valid and usable in optimization.

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2} \sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2} \sqrt{1 - \frac{4\epsilon}{q(x)}}, \quad \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2} \sqrt{1 - \frac{4\epsilon}{q(x)}}, \quad \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with Equation (24) yields the following admissible conditions for the probability ratio $r(x)$:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we select the overlapping region centered at ratio 1. This choice minimizes discrepancy between the two cases and preserves consistency with GRPO's conventional fixed bound, leading to the validity condition below:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we keep the square-root term non-negative by comparing with 0 and using the maximum value, e.g., $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This gives the final **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2} \sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower bounds in Inequality 27, we choose separate hyperparameters ϵ_{high} and ϵ_{low} so the adaptive curve aligns with GRPO at selected reference points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To align dynamic-adaptive clipping with fixed symmetric clipping at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, shown in Figure 4d, we solve Equation (28) for ϵ_{low} and ϵ_{high} .

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \\ q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases} \quad (28)$$

With the common GRPO setting $\epsilon = 0.2$, this alignment gives approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. The inequality $\epsilon_{low} \neq \epsilon_{high}$ serves a different purpose from DAPO’s asymmetric fixed clipping: it calibrates a probability-dependent adaptive curve rather than imposing two global constants.

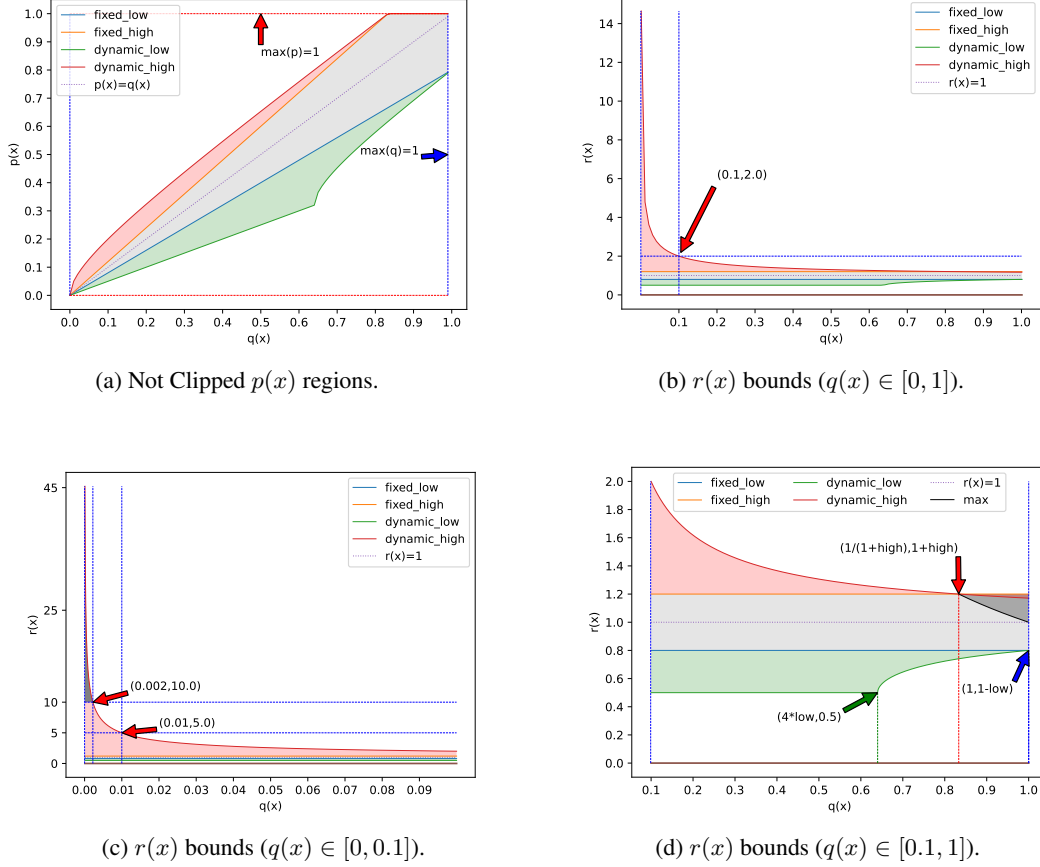


Figure 4: Fixed and dynamic-adaptive clipping bounds. Lines compare fixed clipping ($\epsilon = 0.2$) with dynamic-adaptive clipping ($\epsilon_{low} = 0.16$, $\epsilon_{high} = 0.2$), highlighting the expanded rare-token update region.

Dynamic versus fixed clipping: Figure 4 contrasts GRPO’s fixed clipping interval $[1 - \epsilon, 1 + \epsilon]$ with DCPO’s probability-aware adaptive clipping. For a token with old probability $q(x)$, the fixed bound always provides the same relative width 2ϵ ; therefore, as $q(x)$ decreases, the absolute exploration space shrinks linearly (Figure 4a). Dynamic-adaptive clipping removes this weakness because its relative width scales inversely with $q(x)$, keeping the effective exploration space far more consistent across the probability range.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the fixed interval $[1 - \epsilon, 1 + \epsilon]$ remains unchanged and therefore fails to provide additional room for low-probability tokens. The dynamic-adaptive bound expands rapidly instead (Figure 4b), giving rare tokens the larger relative update space required for efficient RL exploration.

Behavior for extremely low probabilities: Figure 4c focuses on the rare-token regime $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound for $r(x)$ grows rapidly with inverse-square-root behavior, whereas the fixed upper bound remains constant. To prevent instabilities such as gradient explosion or vanishing, we follow the dual-clipping motivation (Ye et al., 2020) and impose a hard ceiling $r_{max} = 10$. The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, where the adaptive upper limit is capped at r_{max} .

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\varepsilon$, dynamic-adaptive clipping gives $dynamic_{low} = 0.5$, while fixed clipping stays at $fixed_{low} = 1 - \varepsilon = 0.8$. As $q(x)$ increases beyond this threshold, the adaptive lower bound contracts smoothly toward the fixed value. The upper bound behaves analogously: it exceeds the fixed bound when $q(x) < \frac{1}{1+\varepsilon_{high}}$ and merges with it for larger $q(x)$. Thus, the adaptive interval $[dynamic_{low}(q(x), \epsilon_{low}), dynamic_{high}(q(x), \epsilon_{high})]$ widens the admissible region for rare tokens while preserving tight fixed-like control for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) includes both positive- and negative-advantage terms. Under the minimum-loss operation, a positive-advantage ratio $r(x)$ is clipped within $[0, dynamic_{high}(q(x), \epsilon_{high})]$, while a negative-advantage ratio uses $[dynamic_{low}(q(x), \epsilon_{low}), \infty)$. Because excessively large $r(x)$ can destabilize training, we cap the ratio for both positive and negative advantages at $r_{max} = 10$. This safeguard follows Ye et al. (2020) and prevents adaptive bounds from creating overly large importance weights.

In short, for both positive and negative advantages, dynamic-adaptive clipping gives low-probability rare tokens more principled exploration space than fixed clipping.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the i -th iteration in which prompts generate responses and update model parameters. The following statistics define the reward normalization used by SAS.

For response-reward means, Equation (29) defines the current-step, historical, and cumulative quantities:

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

For response-reward variances, Equation (30) defines the corresponding current-step, historical, and cumulative quantities:

$$\begin{cases} \sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\ \sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{old}^i)^2} \\ \sigma_{total}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^i)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)} \end{cases} \quad (30)$$

To reduce resource usage when computing σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we use equivalent updates from μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i rather than repeatedly recomputing statistics from all original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments use the following prompt template, based on the Qwen-Math template (Yang et al., 2024b), to keep training and evaluation formatting consistent.

Qwen-Math Template

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant
```

A.9 TRAINING SETTING

We set GRPO’s symmetric clipping threshold to $\epsilon = 0.2$. For DAPO, we use lower/upper clipping thresholds $(0.2, 0.28)$, an additional 512-token soft-punish cache, and maximum probability ratio 10 for negative advantages. For GSPO, we use lower/upper clipping thresholds $(3e - 4, 4e - 4)$. For DCPO, we use low/high clipping thresholds $(0.16, 0.2)$, preserving the GRPO clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1 + \epsilon)$ and $(q(x), r(x)) = (1, 1 - \epsilon)$ as shown in Figure 4d; unlike DAPO, DCPO sets the maximum probability ratio to 10 for both positive and negative advantages.

We train for 400 steps with response-generation batch size 512 and parameter-update mini-batch size 32. Sampling uses temperature 1 and top_p 1 to generate $G=16$ responses per problem. The prompt maximum length is 1,024 and the response maximum length is 3,072, substantially shorter than the 20k maximum length used in the original DAPO.

We adapt the training codebase from Verl (Sheng et al., 2024) and run experiments on 32xH20 GPUs. Unless stated otherwise, hyperparameters match DAPO. For loss computation, we retain the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation from GRPO or DAPO rather than Verl’s approximate weighted calculation, ensuring compatibility across different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports average Avg@1, computed by averaging greedy-decoding results over MATH500, AMC23, AIME24, and AIME25. DCPO consistently outperforms GRPO and DAPO across all four models and outperforms GSPO on the two models where GSPO is evaluated. Its trajectory is also more stable, avoiding the severe drop observed in GRPO and the large fluctuations observed in DAPO. These results show that DCPO more reliably strengthens model reasoning during training.

Avg@32 Trend Figure 6 compares Avg@32 for DCPO, GRPO, DAPO, and, where available, GSPO across model sizes on AMC23, AIME24, and AIME25. The sampled-generation curves clearly show DCPO’s robustness and stability. GRPO exhibits large fluctuations and occasional collapse; on Qwen2.5-14B, it drops sharply after roughly 170 training steps. Across all evaluated models from 1.5B to 14B, DCPO follows a consistently upward trajectory with minimal fluctuation, including in early training where it grows fastest. Compared with DAPO, DCPO achieves stronger performance while using roughly half the training time and resources. These results demonstrate that dynamic-adaptive clipping and smooth advantage standardization jointly improve response-level exploration and label-level diversity.

Entropy Trend Figure 7 shows that GRPO undergoes sharp entropy decay after an initial warm-up and quickly settles into a flat low-entropy regime. This entropy collapse indicates rapid loss of policy randomness, reduced sampled-response diversity, and limited room for later policy optimization.

DAPO and DCPO show model-dependent entropy behavior. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining trajectories. On the larger base

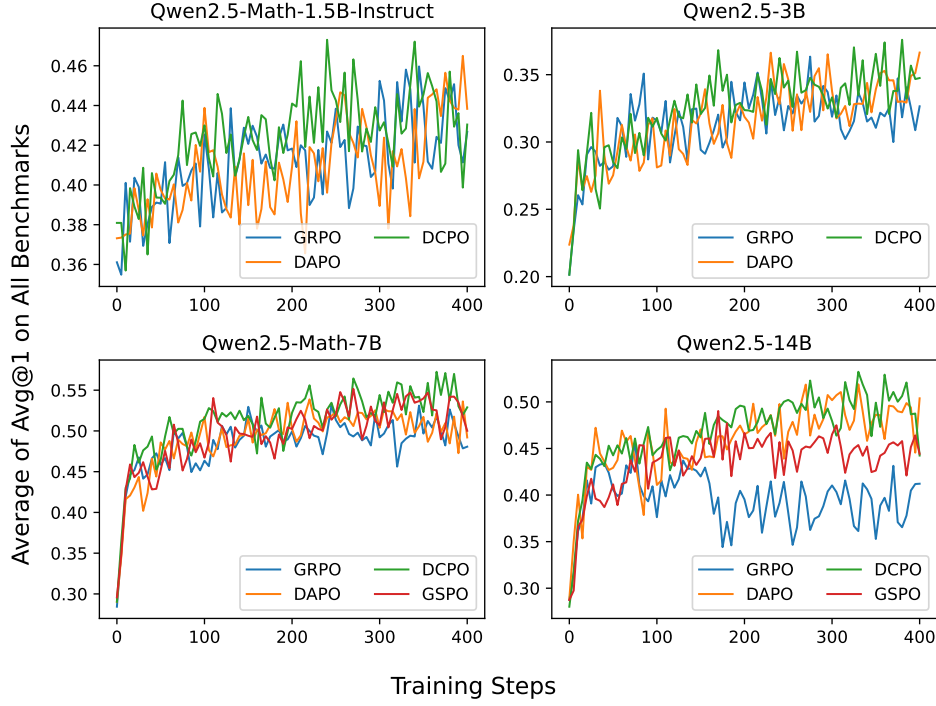


Figure 5: Avg@1 benchmark performance, showing DCPO’s consistent greedy-decoding gains.

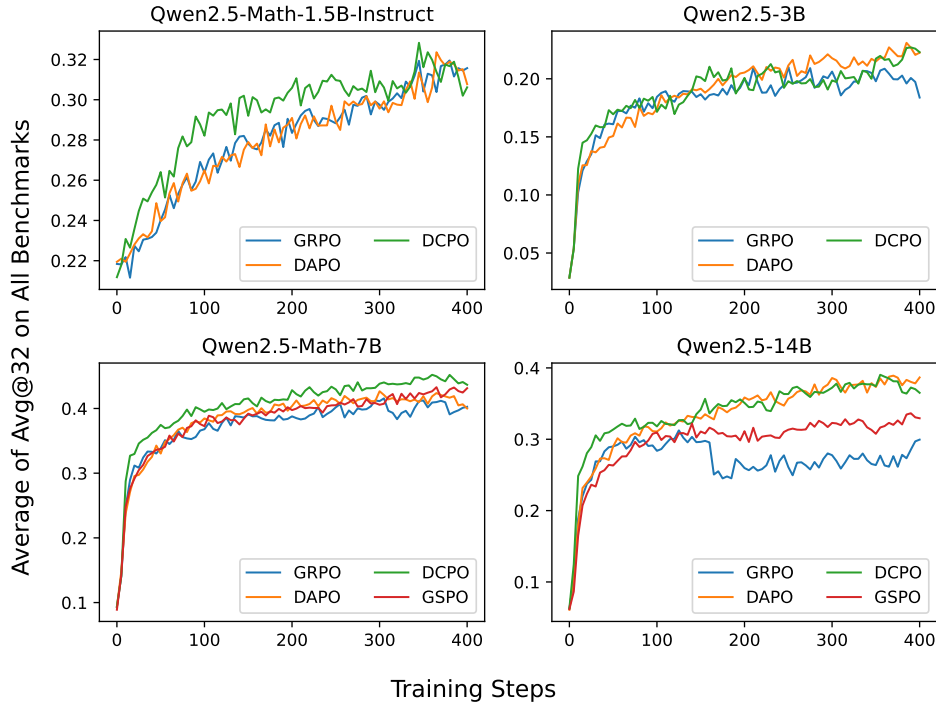


Figure 6: Avg@32 benchmark performance, highlighting DCPO’s stronger sampled-generation robustness.

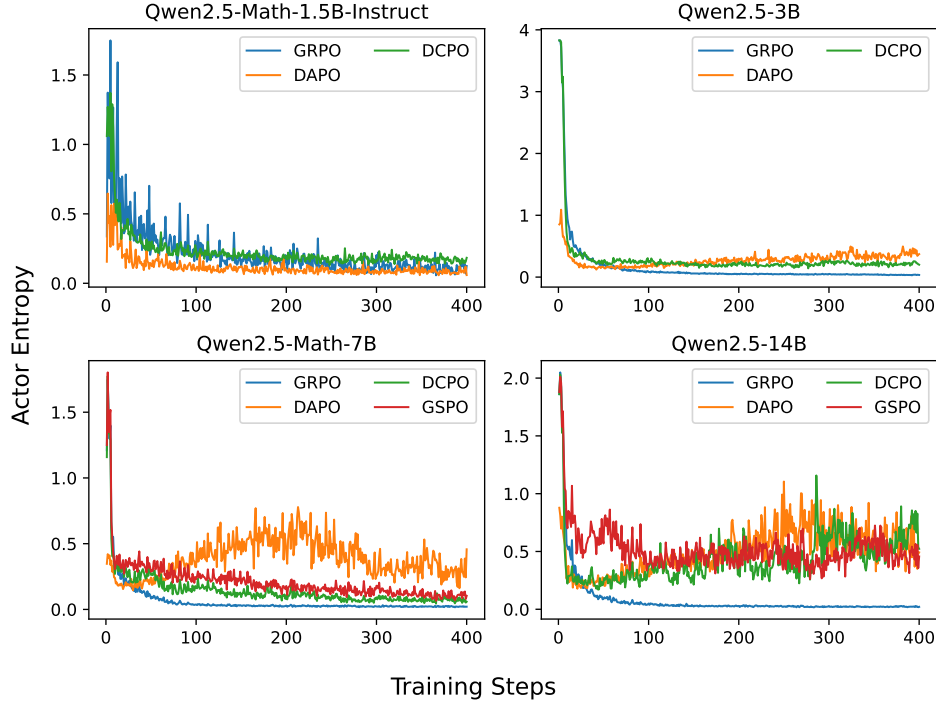


Figure 7: Model entropy during training, showing DCPO’s balanced exploration regime.

models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), the methods separate clearly. DAPO maintains the highest entropy with a rising but highly fluctuating curve. On Qwen2.5-Math-7B, GSPO’s entropy is close to DCPO’s and its average metric performance is also comparable; on Qwen2.5-14B, GSPO instead follows a fluctuating downward performance trend. DCPO occupies the desired middle regime: its entropy curve is smoother and remains bounded between GRPO’s low-entropy collapse region and DAPO’s high-variance region.

These observations support a U-shaped relationship between entropy and training performance. Too little entropy removes exploratory signal and induces premature convergence, whereas excessive or unstable entropy undermines training stability. DCPO maintains entropy in a moderate, well-balanced band that supports both convergence and exploration, with the benefit becoming especially clear as model capacity increases. This entropy behavior helps explain DCPO’s stronger inference performance and more robust policy learning across model sizes.